

Assignment 2

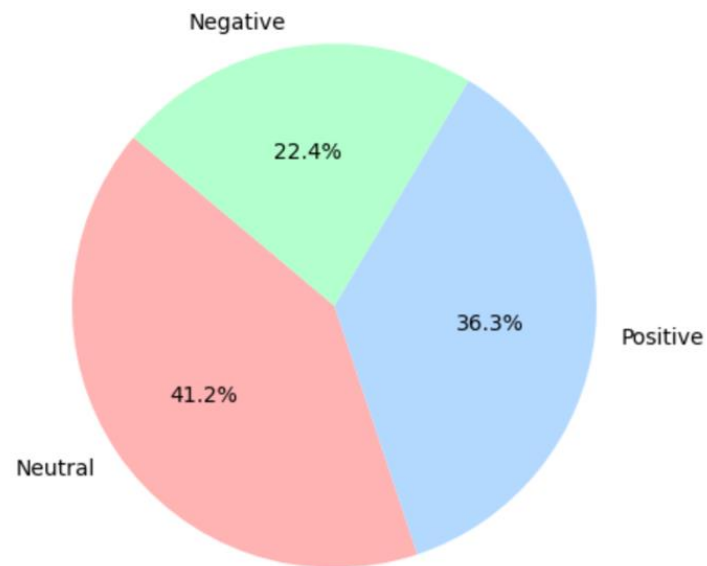
Sentiment Analysis of Public Health Tweets for Early Disease Outbreak Detection

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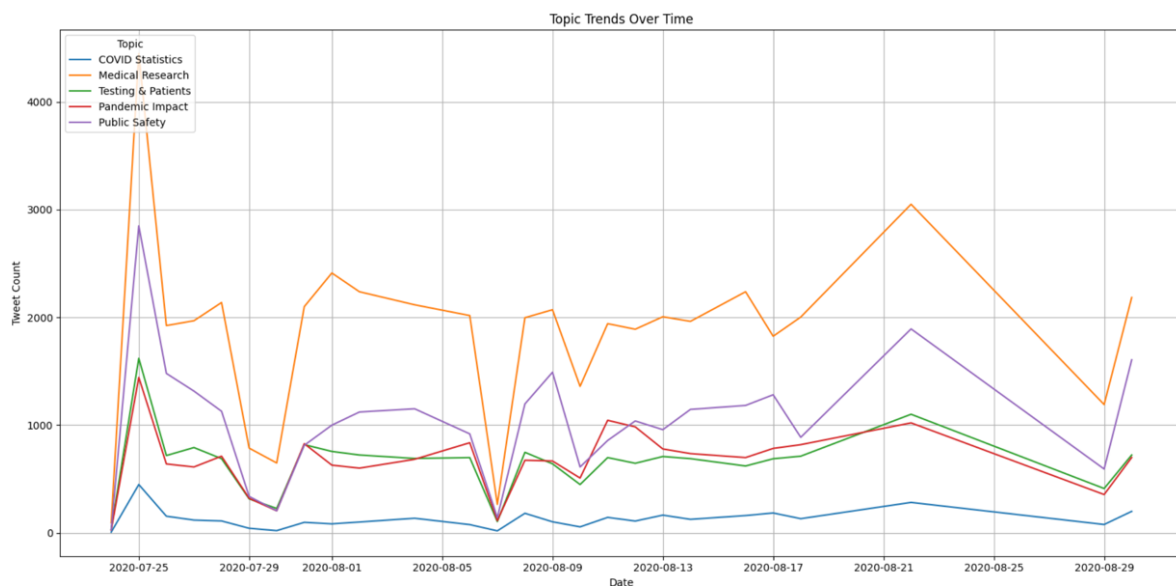
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➤ Sentiment Distribution



• LDA Topic Trends Over Time



- **Key Insight**

- In the sentiment analysis, most tweets were classified as neutral (41.2%), followed by positive (36.3%), and negative (22.4%), indicating that public conversation leaned toward informational or balanced tone rather than emotional extremes.
- The three most frequent words in the dataset were “covid”, “case”, and “new”, suggesting that the public discourse was largely centred on pandemic updates and newly reported cases.
- Through LDA topic modelling, we extracted five dominant themes, which included:
 - COVID Statistics (e.g., covid, case, death),
 - Healthcare System & Outcomes (e.g., health, scientist, worker),
 - Testing & Prevention (e.g., test, hospital, patient),
 - Pandemic Impact & Awareness (e.g., work, world, impact),
 - Safety Practices (e.g., mask, people, wear).

These topics highlight that public discussion spanned both factual data reporting and social concerns, with a strong focus on health measures, testing, and the societal consequences of the pandemic.

- **Ethical Considerations in Social Media Mining**

- **User Privacy:** Even though the data is public, researchers must anonymize content and avoid exposing individual identities. Additionally, users may unintentionally share personal information about others without their consent, which further complicates privacy risks.
- **Information Reliability:** Content shared on social media may not always be accurate or verified, which poses a risk of spreading misinformation if not carefully filtered.
- **Data Bias & Representativeness:** Social media data may not reflect the broader population, leading to biased insights. For example, in our dataset, some users used trending hashtags like #covid19 even when their tweet content was unrelated, simply to gain visibility. This misalignment can distort topic modeling and hashtag-based analysis.